

AWS CAPSTONE PROJECT — Group 10

Team Members

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- Nicholina Adjetey
- Emmanuel Yeboah
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Project Summary

This project is a hands-on cloud infrastructure capstone completed by Group 10. The goal is to plan, build, secure, automate, and monitor a fully functional web application deployment on AWS, working entirely within Free Tier limits.

The web application deployed through this infrastructure is Ghana Political Pulse, a static HTML, CSS, and JavaScript news platform covering Ghanaian politics, elections, and government activities.

The project runs across four phases. Phase 1 establishes the foundation with IAM configuration, CLI setup, EC2 and S3 provisioning, and ACM certificate request. Phase 2 covers web application development and deployment behind an Application Load Balancer. Phase 3 hardens security through CloudFront integration, HTTPS enforcement, and S3 Origin Access Control. Phase 4 completes the project with a GitHub Actions CI/CD pipeline, CloudWatch monitoring, and AWS Budgets cost alerts.

All infrastructure runs on a t3.micro EC2 instance named Group10-capstone-server with static assets stored in an S3 bucket named group10-capstone-bucket. The team of five manages all code and documentation through GitHub with peer-reviewed pull requests, and tracks progress on a Trello board.

Project Objectives

The objectives of this project are to:

- Design and deploy a production-grade web application on AWS cloud infrastructure from the ground up
- Apply Identity and Access Management best practices using the principle of least privilege across all AWS resources

- Build and configure a scalable, fault-tolerant architecture using EC2, ALB, and Auto Scaling
- Secure the application using HTTPS enforcement, CloudFront CDN integration, and Origin Access Control on S3
- Automate application deployments using a GitHub Actions CI/CD pipeline
- Monitor application health and infrastructure performance using CloudWatch alarms and logs
- Manage cloud costs responsibly within AWS Free Tier limits using budget alerts
- Document all infrastructure decisions, configurations, and architecture in a shared GitHub repository

Phase 1: Environment Setup & IAM Configuration Intern Cohort

Phase summary

This phase covers the foundational setup required before application development begins, including AWS account configuration, Identity and Access Management, AWS CLI installation, and the provisioning of core infrastructure resources.

- **IAM Configuration**

Each member is assigned to a phase and is assisted as a group

Permissions are assigned at the group level. No policies are attached directly to individual users. The following least-privilege managed policies were attached to each team member to the phase and AmazonEC2FullAccess is allocated and to cover all four project phases:

- AmazonEC2FullAccess — Phases 1, 2, 3, 4
- AmazonS3FullAccess — Phases 1, 2, 3
- AWSCertificateManagerFullAccess — Phases 1, 3
- CloudFrontFullAccess — Phase 3
- ElasticLoadBalancingFullAccess — Phases 2, 3
- CloudWatchFullAccess — Phase 4
- AutoscalingFullAccess - Phase 2
- IAMFullAccess — Phases 1, 3, 4
- AWSBudgetsActionsWithAWSResourceControlAccess — Phase 4

Each IAM user was provisioned with a console password for AWS Management Console access and an access key and secret key for CLI and programmatic access.

The screenshot shows the AWS IAM console interface. The left sidebar contains navigation links for Dashboard, Access Management (Roles, Policies, IAM users, IAM user groups, Identity providers, Account settings, Root access management, Temporary delegation requests), Access reports (Access Analyzer, Resource analysis), and Storage management and security. The main content area is titled 'IAM users (3)' and includes a search bar, a 'Delete' button, and a 'Create user' button. Below this is a table of IAM users:

	User name	Path	Group	Last activity	MFA	Password age
☐	Barbara	/	1	24 minutes ago	-	12 hours
☐	Nicholina	/	1	12 hours ago	-	12 hours
☐	Susan	/	1	15 minutes ago	-	12 hours

- **S3 buckets**

An S3 bucket named group10-capstone-bucket was created in us-east-1. All public access settings are blocked. Assets will be served through CloudFront using Origin Access Control, which will be configured in Phase 3. Server-side encryption using SSE-S3 is enabled by default.

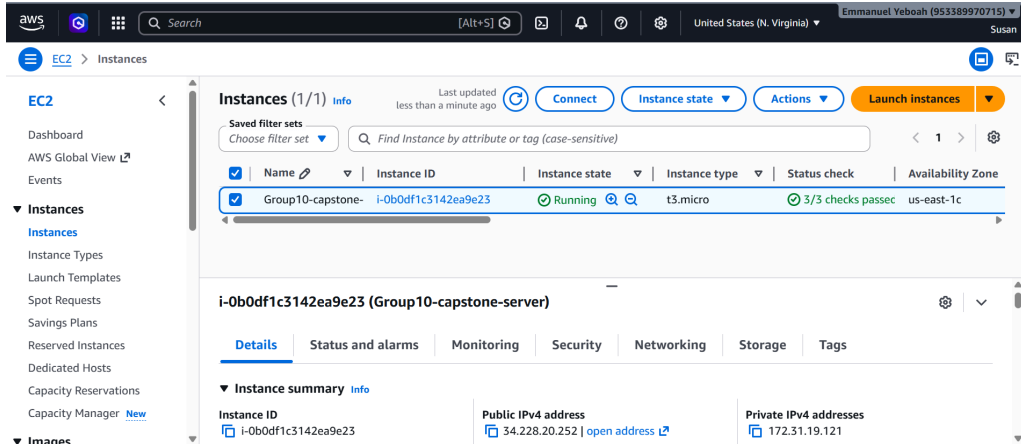
The screenshot shows the AWS S3 console interface. The left sidebar contains navigation links for Amazon S3, Buckets (General purpose buckets, Directory buckets, Table buckets, Vector buckets), Files (File systems), Access management and security (Access Points, Access Points for FSx, Access Grants, IAM Access Analyzer), and Storage management and security. The main content area is titled 'Buckets' and includes tabs for 'General purpose buckets' and 'Directory buckets'. Below this is a table of S3 buckets:

	Name	AWS Region	Creation date
☐	group10-capstone-bucket-953389970715-us-east-1-an	US East (N. Virginia) us-east-1	June 10, 2026, 22:12:45 (UTC+00:00)

Below the table, there are two sections: 'Account snapshot' and 'External access summary', both with 'Updated daily' labels and 'View dashboard' buttons.

- **EC2 Instance**

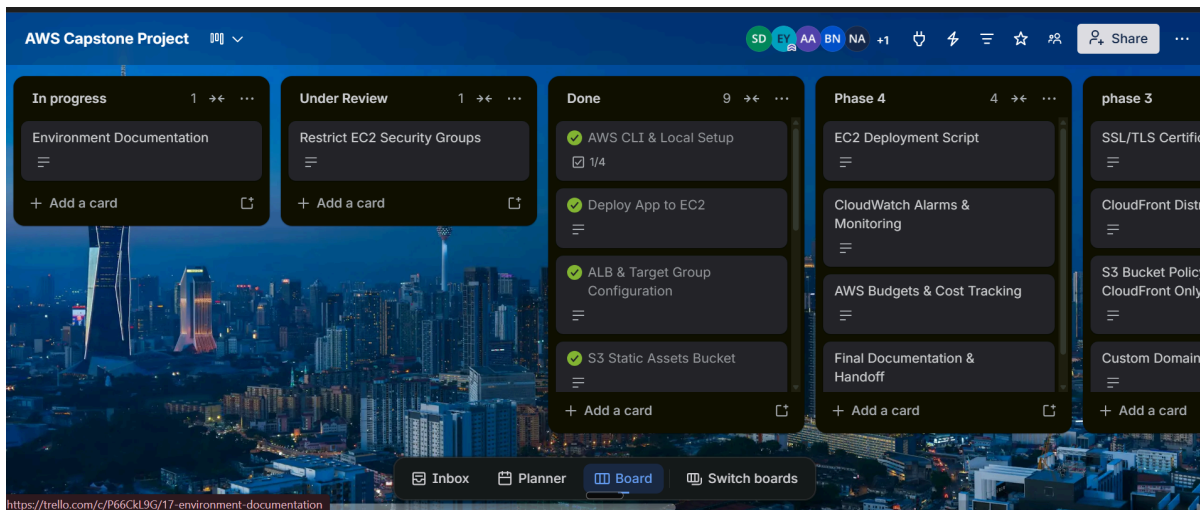
An EC2 instance named Group10-capstone-server was launched successfully and is currently in a Running state with 3/3 status checks passed.

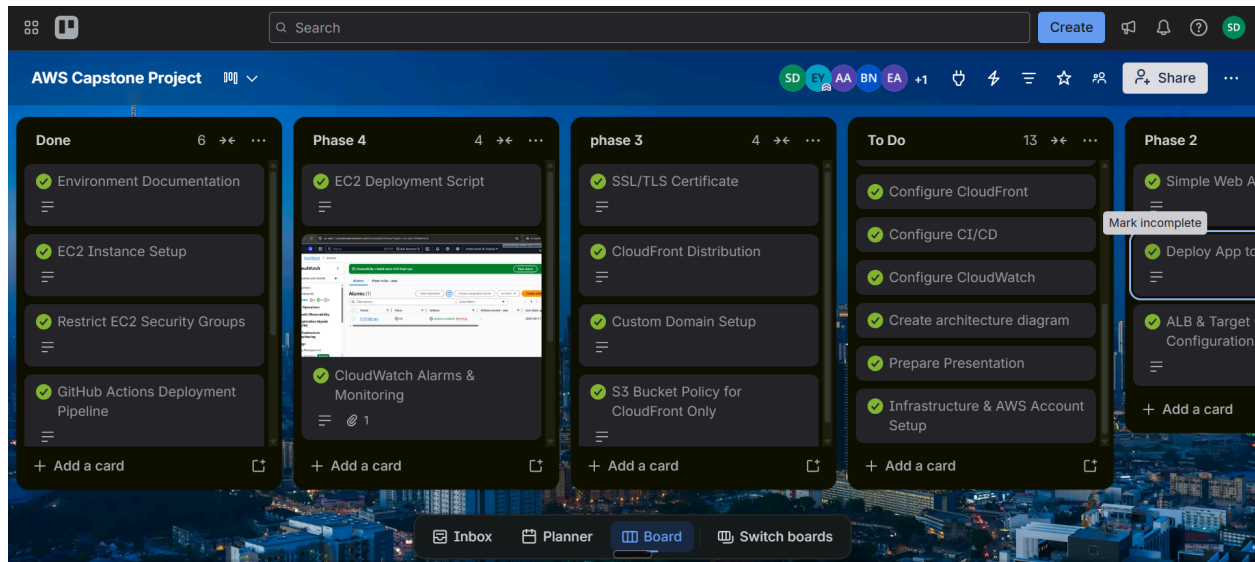


- **Trello**

Set up a Trello board with columns: Backlog, In Progress, Review, Done and add all team members. Find attached the link:

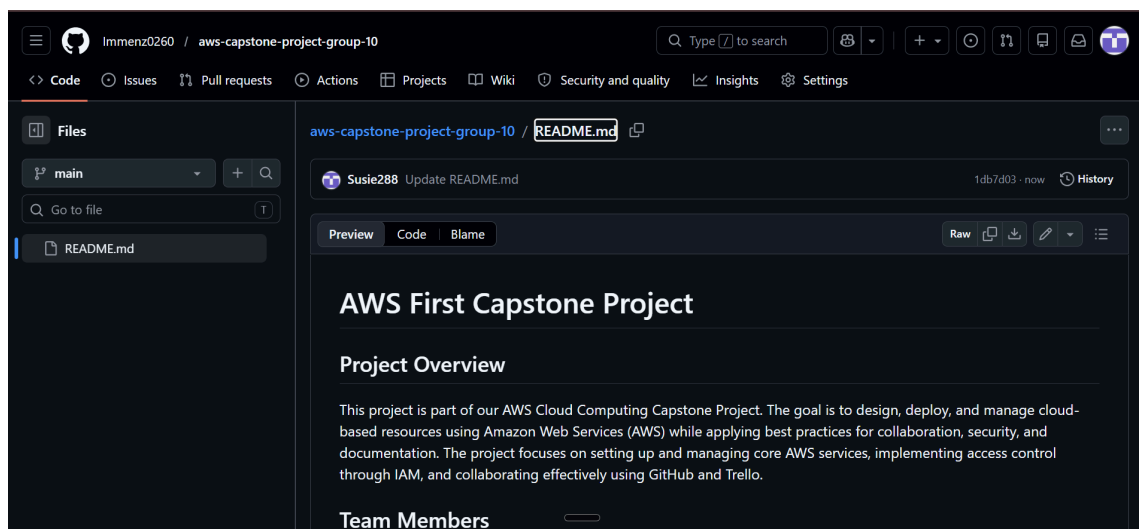
<https://trello.com/invite/b/6a2867d169aac031efe492ea/ATTIc58f62b9cb1147c10bdaef51ebdd7d60DCF953B8/aws-capstone-project>

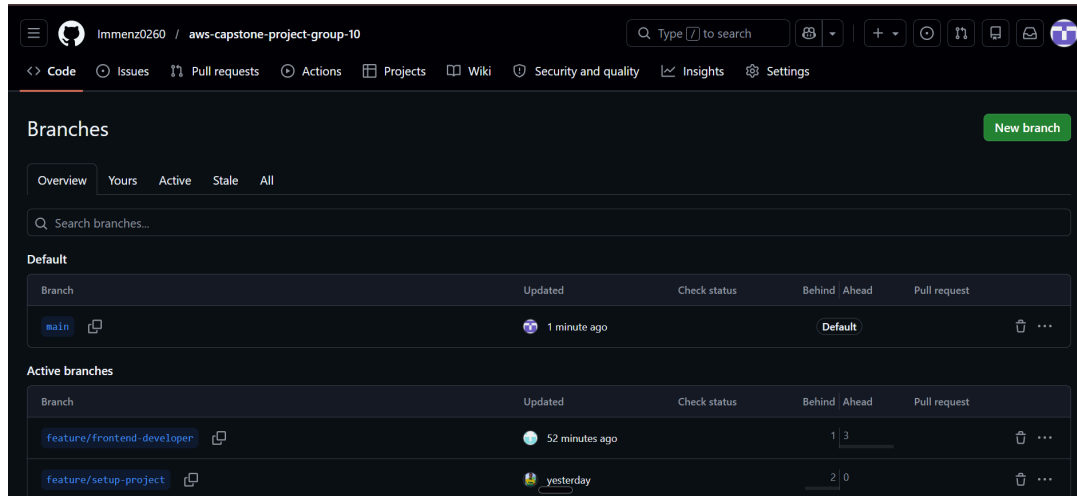




- **GitHub Repository**

A GitHub repository named `aws-capstone-project-group-10` was created under the account `Immenz0260`. The repository contains a `README.md` file titled `AWS First Capstone Project`, which documents the project overview, objectives, and team members. The `main`, `develop`, and `feature` branches are created. Attached is the link to view repo





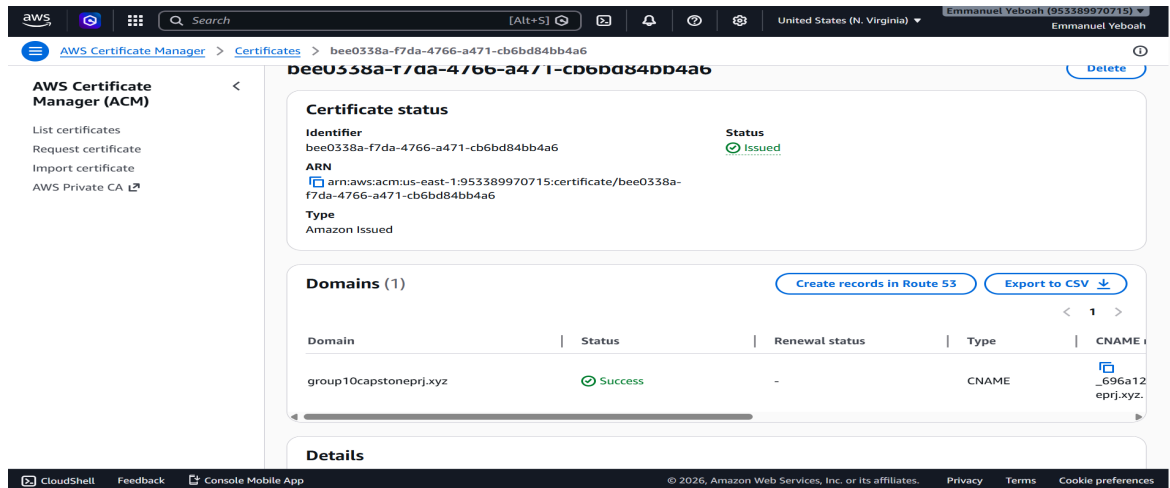
- **AWS CLI install and config**
CLI successfully created and configured

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\keste> aws --version
aws-cli/2.35.1 Python/3.14.5 Windows/11 exe/AMD64
PS C:\Users\keste> aws configure --profile project
AWS Access Key ID [*****AH46]: AKIA536TYOUN6VKQ0FN7
AWS Secret Access Key [*****dcN5]: NUBcBoH62FdFBqAuAax8mcS2IztYv69guXYQtQ87
Default region name [us-east-1]: us-east-1
Default output format [json]: json
PS C:\Users\keste> aws s3 ls --profile project
2026-06-10 22:12:45 group10-capstone-bucket-953389970715-us-east-1-an
```

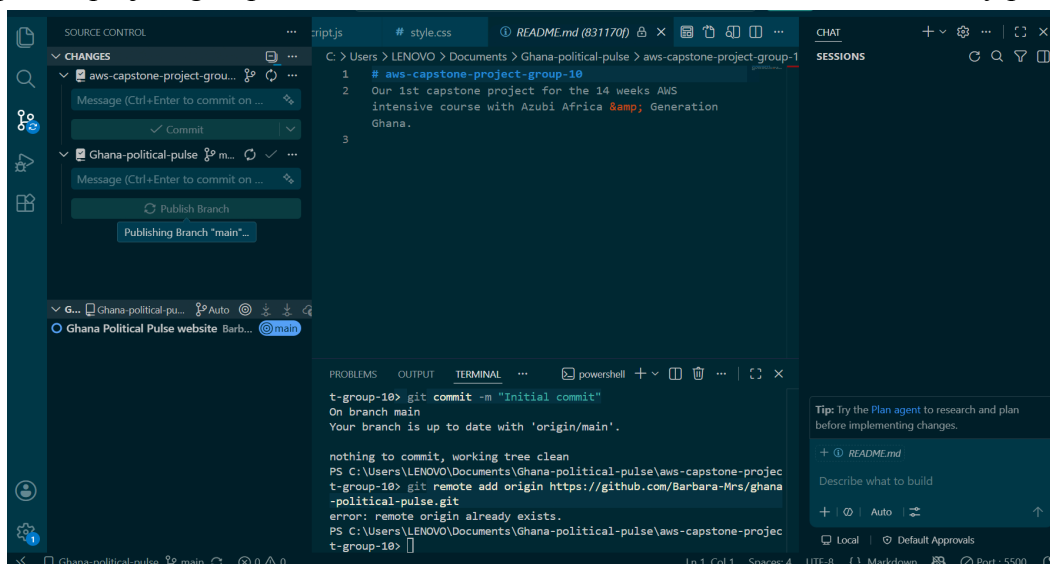
- **ACM certificate DNS validation**
DNS validation completed successfully for the domain **group10capstoneprj.xyz**. The certificate is ready to be attached to CloudFront in Phase 3. The CNAME record was added correctly to the domain DNS settings.



Phase 2: Web Application Development & Deployment

- **Git push**

The project code was pushed to GitHub from VS Code using Git. The repository `aws-capstone-project-group-10` was committed and the main branch was successfully published.



- Development and deployment of web app

Ghana Politics Hub is developed and deployed on an EC2 instance. The app serves as a political news and civic engagement platform for Ghana. Static assets including CSS, JS, and images were uploaded to S3 and served via public URL. The homepage is live and accessible via the ALB endpoint.

```
aws
Search [Alt+S]
United States (N. Virginia) Emmanuel Yeboah (953389970715) Barbara

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Wed Jun 10 22:59:51 2026 from 18.206.107.29
[ec2-user@ip-172-31-19-121 ~]$ sudo dnf update -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-19-121 ~]$ sudo dnf install nginx -y
Last metadata expiration check: 0:00:24 ago on Thu Jun 11 08:09:27 2026.
Dependencies resolved.

Package Architecture Version Repository Size
-----
Installing:
nginx x86_64 1:1.30.2-1.amzn2023.0.1 amazonlinux 34 k

[ec2-user@ip-172-31-19-121 ~]$ sudo dnf install git -y
Last metadata expiration check: 0:05:10 ago on Thu Jun 11 08:09:27 2026.
Dependencies resolved.

Package Architecture Version Repository Size
-----
Installing:
git x86_64 2.50.1-1.amzn2023.0.1 amazonlinux 53 k
Installing dependencies:
git-core x86_64 2.50.1-1.amzn2023.0.1 amazonlinux 4.9 M
git-core-doc noarch 2.50.1-1.amzn2023.0.1 amazonlinux 2.8 M
perl-Error noarch 1:0.17030-2.amzn2023.0.1 amazonlinux 42 k
perl-File-Find noarch 1.37-477.amzn2023.0.9 amazonlinux 26 k
perl-Git noarch 2.50.1-1.amzn2023.0.1 amazonlinux 41 k
perl-TermReadKey x86_64 2.38-9.amzn2023.0.3 amazonlinux 36 k
perl-lib x86_64 0.65-477.amzn2023.0.9 amazonlinux 15 k

Transaction Summary
-----
Install 8 Packages

Total download size: 7.9 M
Installed size: 41 M
Downloading Packages:
```

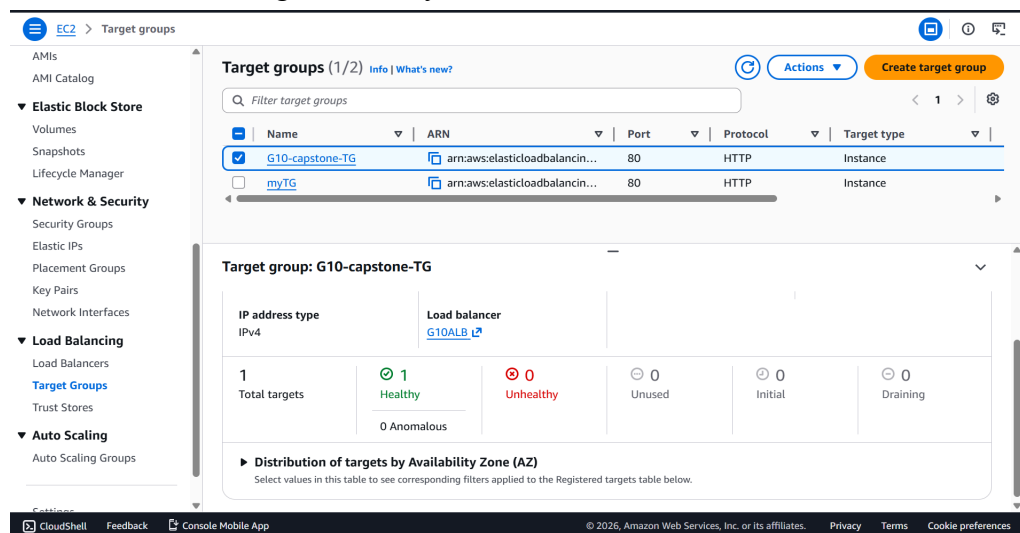
i-0b0df1c3142ea9e23 (Group10-capstone-server)
PublicIPs: 34.228.20.252 PrivateIPs: 172.31.19.121



[Ghana Political Pulse](#)

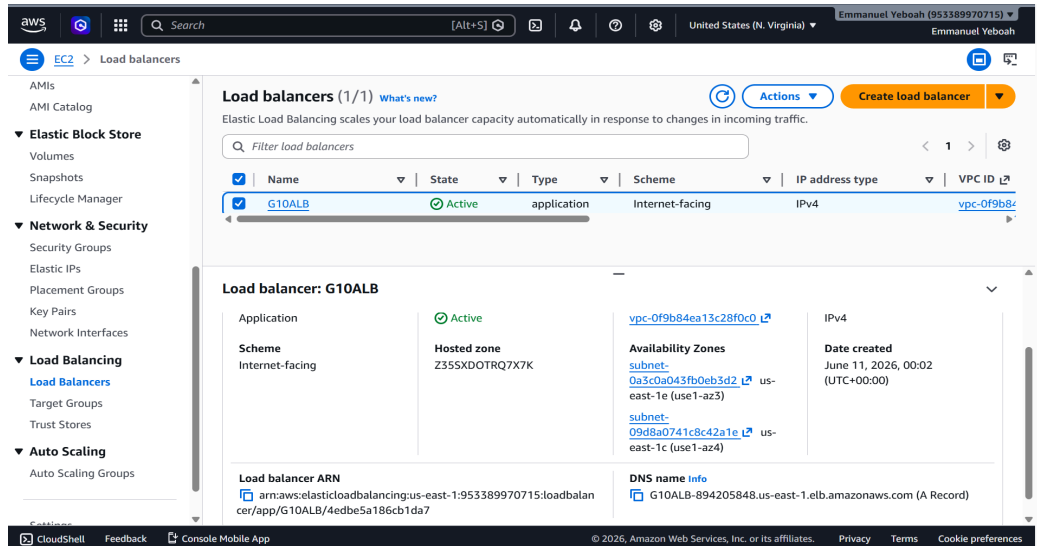
- **Configuring target group**

A Target Group named **G10-capstone-TG** was configured in AWS EC2, pointing to the EC2 instance on port 80 using the HTTP protocol. The target group is linked to the **G10ALB** load balancer and shows 1 total target with a status of **1 Healthy** and 0 Unhealthy. Attached is a screenshot confirming the healthy state.



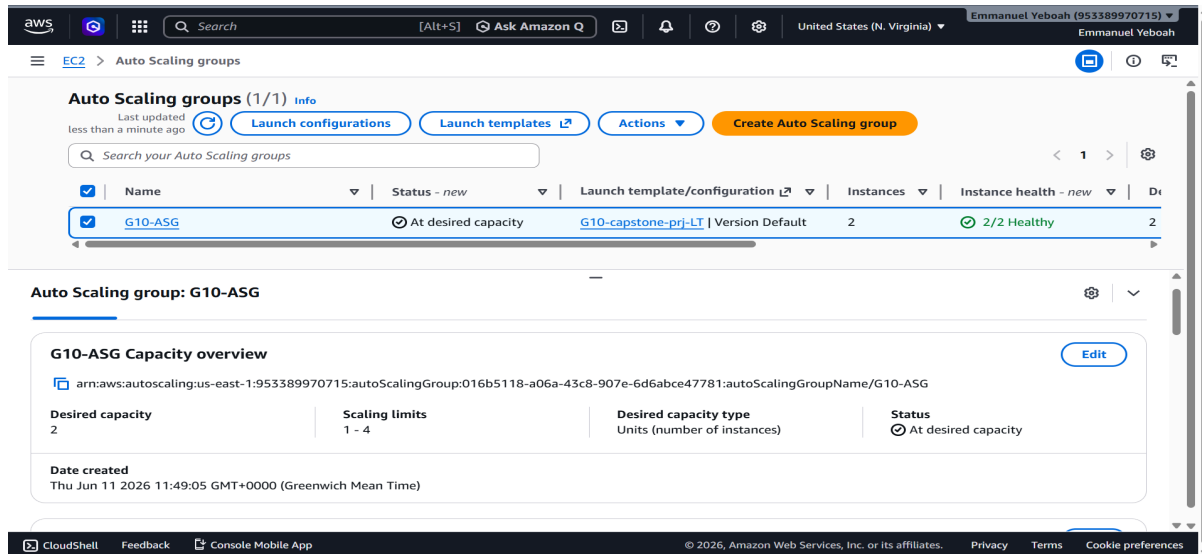
- **Application load balancer**

An Application Load Balancer named G10ALB was successfully created and is currently Active. It is internet-facing and deployed across multiple availability zones.

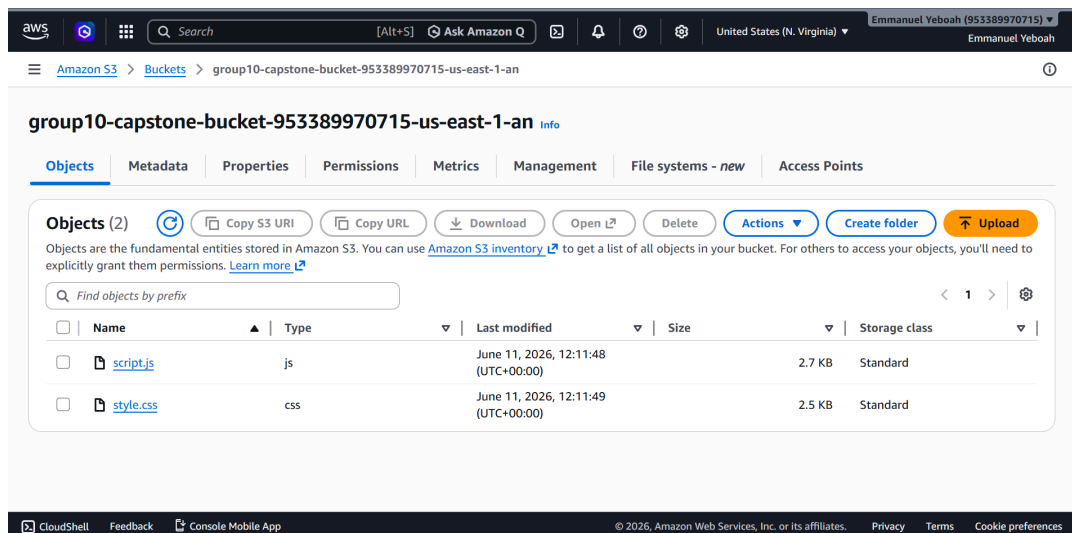


- **Setting up Auto scaling**

An Auto Scaling Group named G10-ASG was successfully set up and is currently at desired capacity with 2/2 instances healthy. Scaling limits are configured between 1 and 4 instances.



- Upload static assets (images, CSS, JS) to S3 and serve via public URL



Phase 3: Security Hardening & CloudFront Integration

Create a CloudFront distribution with ALB as the origin

A CloudFront distribution named cloudfront-group10-distribution was successfully created and is currently deploying. The distribution is configured to use all edge locations for best performance.

CloudFront > Distributions > E1QAVJJUUBK90

Successfully created new distribution.

cloudfront-group10-distribution Standard View metrics

Details

Distribution domain name
d2c3ne5ulpfeqv.cloudfront.net

Billing
Pay-as-you-go
Switch to a plan

ARN
arn:aws:cloudfront:953389970715:distribution/E1QAVJJUUBK90

Last modified
Deploying

General

Security

Origins

Behaviors

Error pages

Invalidations

Logging

Tags

Settings Edit

Name
cloudfront-group10-distribution

Description
Capstoneproject

Price class
Use all edge locations (best performance)

Supported HTTP versions
HTTP/2, HTTP/1.1, HTTP/1.0

Alternate domain names
-
Add domain

Standard logging
[Off](#)

Cookie logging
[Off](#)

Default root object
-

Cache tag header
-

Apply IAM policies using principle of least privilege for all AWS resources

User	Phase	Permissions
User 1	Phase 1 – Setup	AmazonEC2FullAccess, AmazonS3FullAccess, AWSCertificateManagerFullAccess, IAMFullAccess
User 2	Phase 2 – Development	AmazonEC2FullAccess, ElasticLoadBalancingFullAccess, AmazonS3FullAccess, AutoScalingFullAccess
User 3	Phase 3 – Security	CloudFrontFullAccess, AmazonEC2FullAccess, AmazonS3FullAccess, IAMReadOnlyAccess
User 4	Phase 4 – CI/CD & Monitoring	CloudWatchFullAccess, AWSBudgetsFullAccess, AmazonEC2ReadOnlyAccess

IAM > IAM users > Barbara

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

Roles

Policies

IAM users

IAM user groups

Identity providers

Account settings

Root access management

Temporary delegation requests

Access reports

☐

AmazonEC2FullAccess

AWS managed

☐

AmazonS3ExpressFullAccess

AWS managed

☐

CloudWatchFullAccess

AWS managed

Permissions boundary (not set)

Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this user, then customize, create, and attach it to this user. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

Generate policy

No requests to generate a policy in the past 7 days.

IAM > IAM users > Nicholina

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

Roles

Policies

IAM users

IAM user groups

Identity providers

Account settings

Root access management

Temporary delegation requests

Access reports

☐

AmazonEC2FullAccess

AWS managed

Directly

☐

AmazonS3FullAccess

AWS managed

Directly

☐

CloudFrontFullAccess

AWS managed

Directly

☐

IAMReadOnlyAccess

AWS managed

Directly

☐

IAMUserChangePassword

AWS managed

Directly

Permissions boundary (not set)

Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this user, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

Generate policy

No requests to generate a policy in the past 7 days.

IAM > IAM users > Susan

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

Roles

Policies

IAM users

IAM user groups

Identity providers

Account settings

Root access management

Temporary delegation requests

Access reports

☐

AmazonS3FullAccess

AWS managed

Directly

☐

AutoScalingFullAccess

AWS managed

Directly

☐

ElasticLoadBalancingFullA...

AWS managed

Directly

☐

IAMUserChangePassword

AWS managed

Directly

Permissions boundary (not set)

Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this user, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

Generate policy

No requests to generate a policy in the past 7 days.

Enable S3 bucket policies to only allow CloudFront Origin Access Control (OAC)

Bucket: group10-capstone-bucket-953389970715-us-east-1-an

The S3 bucket is configured with Origin Access Control (OAC) to restrict access exclusively to the CloudFront distribution (E1QAVJJUUBK90). Direct public S3 URL access is blocked. The bucket policy allows only the CloudFront service principal with a condition matching the specific distribution ARN.

The screenshot shows the AWS CloudFront console. The left sidebar contains navigation links for CloudFront, Distributions, Policies, Functions, Static IPs, VPC origins, SaaS, Telemetry, and Monitoring. The main content area displays the details for the distribution 'cloudfront-group10-distribution'. A green notification banner at the top states: 'Successfully created origin group10-capstone-bucket-953389970715-us-east-1-an.s3.us-east-1.amazonaws.com.' The distribution details include: Distribution domain name: d2c3ne5ulpfeqv.cloudfront.net, Billing: Pay-as-you-go, ARN: arn:aws:cloudfront::953389970715:distribution/E1QAVJJUUBK90, and Last modified: Deploying. A 'View metrics' button is also visible.

The screenshot shows the AWS Amazon S3 console. The left sidebar contains navigation links for Amazon S3, Buckets, Files, Access management and security, Storage management and insights, and AWS Marketplace for S3. The main content area displays the bucket policy for the bucket 'group10-capstone-bucket-953389970715-us-east-1-an'. A green notification banner at the top states: 'Successfully edited bucket policy.' The bucket policy is shown in JSON format, allowing access to the bucket only for the CloudFront service principal. The policy text is as follows:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::group10-capstone-bucket-953389970715-us-east-1-an/*"
    },
    {
      "Sid": "AllowCloudFrontServicePrincipal",
      "Effect": "Allow",
      "Principal": {
        "Service": "cloudfront.amazonaws.com"
      },
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::group10-capstone-bucket-953389970715-us-east-1-an/*",
      "Condition": {
        "StringEquals": {

```

Security Documentation

Port	Protocol	Direction	Source / Destination	Purpose
80	TCP/HTTP	Inbound	pl-3b927c52 (CloudFront)	Receive traffic from CloudFront only
All	All	Outbound	0.0.0.0/0	Default outbound — allow all

Rationale: The ALB only accepts inbound HTTP traffic originating from CloudFront's managed prefix list (pl-3b927c52), which contains all CloudFront edge node IP ranges. Direct public access to the ALB is blocked.

1.2 EC2 Security Group (launch-wizard-1)

Port	Protocol	Direction	Source / Destination	Purpose
80	TCP/HTTP	Inbound	ALB Security Group ID	Receive traffic from ALB only
22	TCP/SSH	Inbound	AWS EC2 Instance Connect	Console SSH access for administration
All	All	Outbound	0.0.0.0/0	Default outbound — allow all

Rationale: EC2 only accepts HTTP traffic from the ALB security group, ensuring no direct public access to the instance. SSH is available via AWS EC2 Instance Connect through the console only.

aws [Search] [Option+S] United States (N. Virginia) Nicholina

EC2 > Security Groups > sg-0bd5fc7f451a348d2 - launch-wizard-1

✓ Inbound security group rules successfully modified on security group (sg-0bd5fc7f451a348d2 | launch-wizard-1)
 ▶ Details

sg-0bd5fc7f451a348d2 - launch-wizard-1 Actions

Details

Security group name launch-wizard-1	Security group ID sg-0bd5fc7f451a348d2	Description launch-wizard-1 created 2026-06-10 T22:09:22.386Z	VPC ID vpc-0f9b84ea13c28f0c0
Owner 953389970715	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules Outbound rules Sharing VPC associations Related resources - new Tags

Inbound rules (1) Manage tags Edit inbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-0333ae8f93f1351a0	-	HTTP	TCP	80

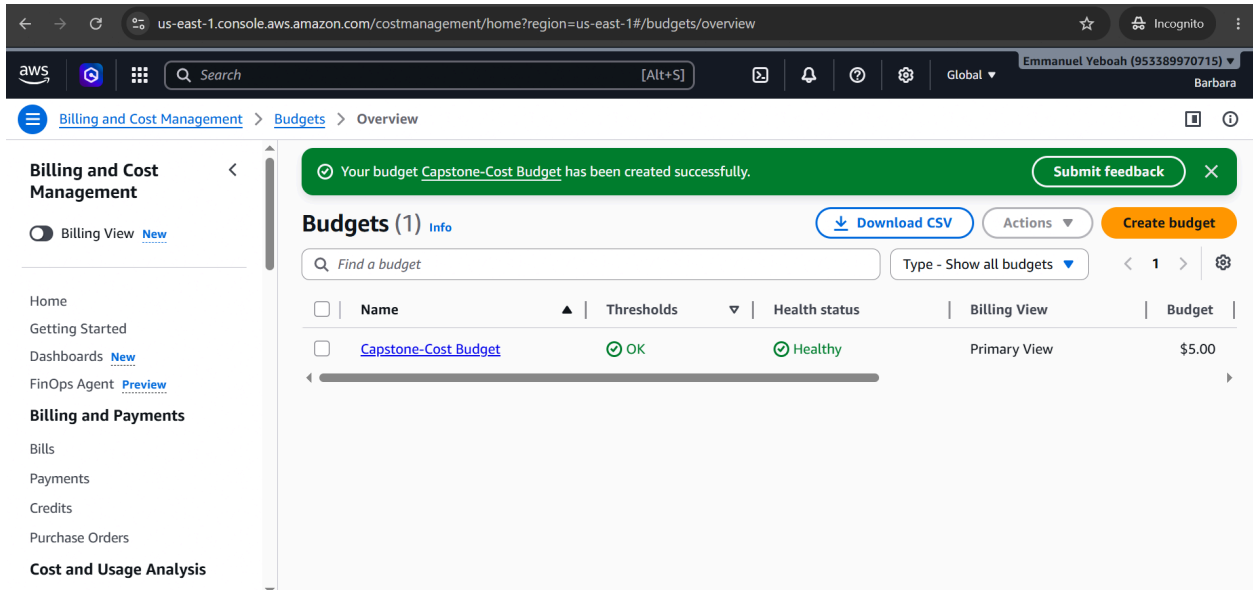
2. CloudFront Configuration

Setting	Value
Viewer Protocol Policy	Redirect HTTP to HTTPS
Origin Protocol Policy	HTTP Only (CloudFront to ALB over AWS internal network)
Cache Policy	Managed-CachingOptimized
Origin	G10ALB (Application Load Balancer)
ACM Certificate	Attached to CloudFront for HTTPS on custom domain

Phase 4: CI/CD Pipeline, Monitoring & Cost Management

Set up AWS Budgets alert to notify when costs approach Free Tier limit

The Capstone-Cost Budget was successfully created at \$5.00 with a Healthy status.



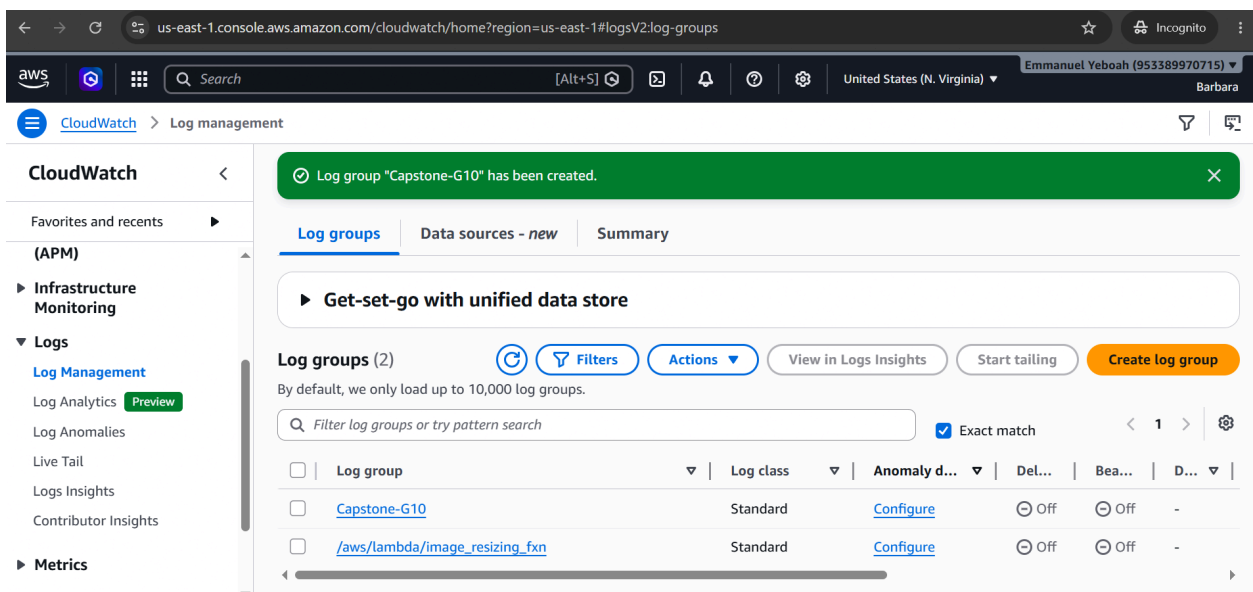
The screenshot shows the AWS Budgets console in the us-east-1 region. A green notification bar at the top states: "Your budget Capstone-Cost Budget has been created successfully." Below this, the "Budgets (1)" section shows a table with one budget:

	Name	Thresholds	Health status	Billing View	Budget
☐	Capstone-Cost Budget	OK	Healthy	Primary View	\$5.00

The left sidebar shows the "Billing and Cost Management" menu with options like Billing View, Home, Getting Started, Dashboards, FinOps Agent, Billing and Payments, and Cost and Usage Analysis.

CloudWatch Alarm (G10-high-cpu)

The alarm **G10-high-cpu** was successfully created and is currently in an OK state with actions enabled.



The screenshot shows the AWS CloudWatch console in the us-east-1 region. A green notification bar at the top states: "Log group 'Capstone-G10' has been created." Below this, the "Log groups (2)" section shows a table with two log groups:

	Log group	Log class	Anomaly d...	Del...	Bea...	D...
☐	Capstone-G10	Standard	Configure	Off	Off	-
☐	/aws/lambda/image_resizing_fxn	Standard	Configure	Off	Off	-

The left sidebar shows the "CloudWatch" menu with options like Favorites and recents, (APM), Infrastructure Monitoring, Logs, Log Management, Log Analytics, Log Anomalies, Live Tail, Logs Insights, Contributor Insights, and Metrics.

Set up a deployment script that SSHs into EC2 and pulls latest code from GitHub

```
Code Blame
1  name: Deploy to EC2
2
3  on:
4
5  push:
6
7    branches:
8
9      - main
10
11 jobs:
12
13   deploy:
14
15     runs-on: ubuntu-latest
16
17     steps:
18
19       - name: Checkout code
20
21         uses: actions/checkout@v4
22
23       - name: Deploy to EC2 via SSH
24
25         uses: appleboy/ssh-action@master
```

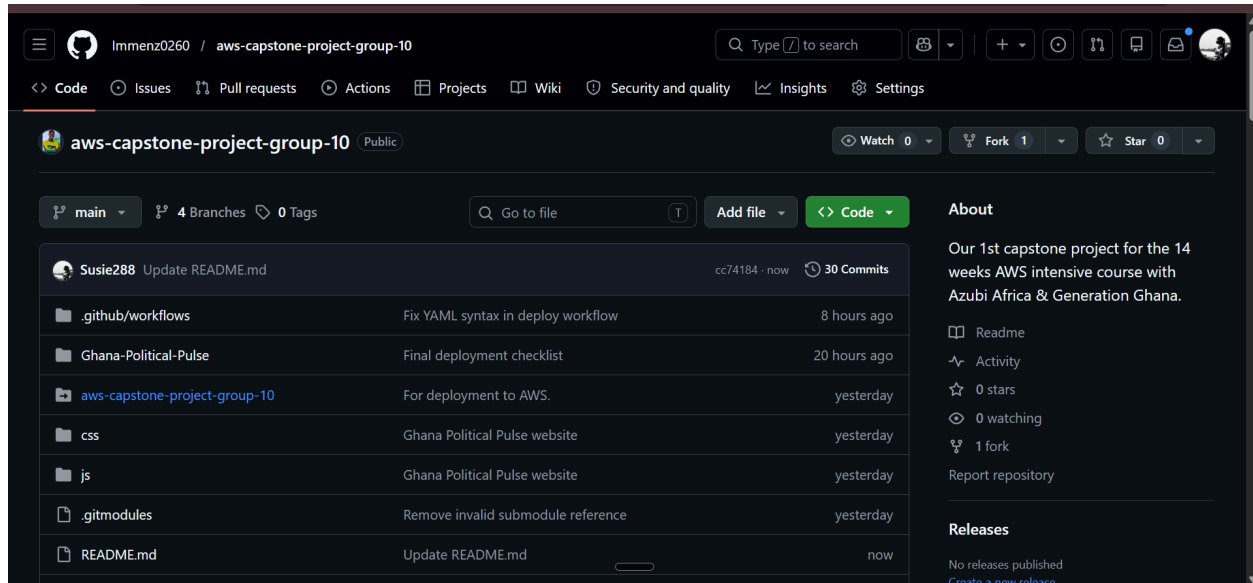
Update Trello board to reflect completed tasks and close out backlog items

The screenshot shows a Trello board for the 'AWS Capstone Project'. The board is organized into five columns representing different stages of the project:

- Done (6 items):** Environment Documentation, EC2 Instance Setup, Restrict EC2 Security Groups, GitHub Actions Deployment Pipeline.
- Phase 4 (4 items):** EC2 Deployment Script, CloudWatch Alarms & Monitoring.
- phase 3 (4 items):** SSL/TLS Certificate, CloudFront Distribution, Custom Domain Setup, S3 Bucket Policy for CloudFront Only.
- To Do (13 items):** Configure CloudFront, Configure CI/CD, Configure CloudWatch, Create architecture diagram, Prepare Presentation, Infrastructure & AWS Account Setup.
- Phase 2 (4 items):** Simple Web A, Deploy App to, ALB & Target Configuration.

Each task card has a green checkmark indicating completion. The 'To Do' column has a 'Mark incomplete' button. The board includes a search bar, a 'Create' button, and a 'Share' button. The bottom navigation bar shows 'Inbox', 'Planner', 'Board', and 'Switch boards'.

Write a final deployment checklist and update the project README with architecture diagram



Links to Resources and Documentation

Amazon CloudFront

<https://us-east-1.console.aws.amazon.com/cloudfront/v4/home?region=us-east-1#/distributions/E1QAVJJUUBK90/security>

Application Load Balancer

<https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LoadBalancers:>

Amazon EC2

<https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-0b0df1c3142ea9e23>

AWS Certificate Manager

[bee0338a-f7da-4766-a471-cb6bd84bb4a6](https://us-east-1.console.aws.amazon.com/certificate-manager/home?region=us-east-1#/certificates/bee0338a-f7da-4766-a471-cb6bd84bb4a6)

AWS IAM

<https://us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/home>

Amazon S3

<https://us-east-1.console.aws.amazon.com/s3/home?region=us-east-1#>

GitHub Actions

<https://github.com/Immenz0260/aws-capstone-project-group-10.git>

Trello Guide

<https://trello.com/invite/b/6a2867d169aac031efe492ea/ATTIc58f62b9cb1147c10bdaef51ebdd7d60DCF953B8/aws-capstone-project>

AWS Budgets

<https://us-east-1.console.aws.amazon.com/costmanagement/home?region=us-east-1#/budgets/details?name=Capstone-Cost%20Budget>

AWS Free Tier

<https://us-east-1.console.aws.amazon.com/costmanagement/home?region=us-east-1#/freetier>